

TEACHER'S LESSON PLAN

Butterfly (WBBPE) · Class V English · Lesson 12 — **Beyond Barriers** (Rakesh Sharma & Kalpana Chawla) · studymatic.in

Class	V (WBBPE)	Subject	English (Butterfly)
Lesson	12 — Beyond Barriers (Indians in space)	Periods	5 × 40 minutes (suggested)
Theme	Dreams, courage and crossing barriers	Skills	Reading, vocabulary, grammar (present continuous), writing

Learning objectives

- Students retell the achievements of Rakesh Sharma and Kalpana Chawla in order.
- Students recall key facts about both astronauts (places, years, shuttles, missions).
- Students identify firsts in space (first man, woman, spacewalk, animal) from a table.
- Students use Word Trove words (miserable, cadet, space shuttle, ruffled, mission, crew).
- Students recognise and use the present continuous tense (am/is/are + verb-ing).

Materials

- Butterfly textbook (pages 121–132); the Earth-from-space and astronaut pictures.
- A globe and a picture of a space shuttle; a world map to mark Patiala, Karnal, Texas and California.
- Two board fact-files (Rakesh Sharma | Kalpana Chawla) to fill in as you read.
- StudyMatic chapter page for notes and the instant quiz: studymatic.in → WBBPE → Class V → English → Lesson 12.

Period-wise teaching plan

Period	Focus	Teacher activity	Student activity
1	Warm-up + the frame	Show the Earth-from-space picture (Let's begin): what do you see? what does blue signify? Set up Barun's sick Sunday and the book. Teach: miserable. Read the start.	Answer Let's begin; predict what the book is about; read the opening.
2	Rakesh Sharma	Read Rakesh Sharma's story (Patiala 1949; NDA cadet 1966; Soyuz T-11, 3 April 1984; 7 days 21 hrs 40 min; 'Sare Jahan Se Achcha'). Teach: cadet, space shuttle. Fill the board fact-file.	Activity 1 (facts about Rakesh Sharma), Activity 2 (T/F); fill the fact-file.
3	Kalpana Chawla	Read Kalpana's story (Karnal 1962, 'Montu'; Aeronautical Engineering; USA 1982; NASA 1988; Columbia 1997 & 2003; named on Mars/a star). Teach: ruffled, mission, crew.	Activity 4 (find answers), Activity 6 (complete the sentences); fill the second fact-file.
4	Vocabulary + space firsts	Sort names into person/place/things; read the space-firsts table (Gagarin, Tereshkova, Leonov, Savitskaya, Laika) and model turning rows into sentences.	Activity 5 (name table), Activity 7 (five sentences from the firsts table).
5	Grammar: present continuous + writing	Teach the present continuous (am/is/are + -ing) from 'I am reading...'; uses — now, around now, planned future. Guide Activity 9 (six sentences on the night sky). Recite 'Zoom, zoom, zoom'.	Activity 8 (present continuous fill-ins), Activity 9 (night-sky paragraph); attempt the StudyMatic 10-question quiz.

Board summary (write-up for the class)

Sick Barun misses a picnic; his mother's **book on India's space travels** cheers him up. **Rakesh Sharma** — first Indian in space — born 1949, Patiala; NDA cadet 1966; **3 April 1984** aboard **Soyuz T-11**, 7 days 21 hrs 40 min; told Indira Gandhi India looked '**Sare Jahan Se Achcha**'. **Kalpana Chawla** — first Indian woman in space — born 1962, Karnal ('Montu'); studied Aeronautical Engineering; USA 1982; NASA 1988; flew on **Columbia** in 1997 (spoke to PM Gujral from orbit) and again on **16 Jan 2003** — Columbia lost with its crew on re-entry, **1 Feb 2003**. A hill on Mars and a star are named after her. **Theme:** hard work and courage carry us 'beyond barriers'.

Word Trove (teach with sentences)

Word	Meaning
miserable	very sad
cadet	a trainee
space shuttle	a vehicle sent to space
ruffled	made the hair uneven
mission	a task
crew	a team

Space firsts (Activity 7)

First	Name
First Man in Space	Yuri Gagarin
First Woman in Space	Valentina Tereshkova
First Man to Walk in Space	Alexei Leonov
First Woman to Walk in Space	Svetlana Savitskaya
First Animal in Space	Laika, the dog

Answer key — textbook activities

- **Activity 1 (Rakesh Sharma):** (a) Patiala, Punjab; (b) 1949; (c) St. George's Grammar School & Kendriya Vidyalaya Tirumalagiri, Hyderabad; (d) July 1966; (e) 7 days, 21 hours, 40 minutes.
- **Activity 2 (T/F):** (a) T; (b) T (Soyuz T-11, 1985 **X** — note: he flew in 1984, so 'went into space in 1985' is F); (c) T; (d) T; (e) T.
- **Activity 3 (T/F):** (a) T; (b) F (1984, not 1985); (c) T; (d) T; (e) T.
- **Activity 4:** (a) Karnal, Haryana; (b) the USA (University of Texas at Arlington); (c) 'Montu'; (d) Columbia; (e) 1st February 2003; (f) a hill on Mars and a star in deep space.
- **Activity 5 (names):** Person — Pinky, Ravi, Rakesh, Kalpana, Barun, Osman; Place — India, USA, Haryana, California; Things — Soyuz T-11, Columbia.
- **Activity 6:** (1) Rakesh Sharma; (2) the first Indian woman to go to space; (3) the value of hard work and sincerity; (4) Mr. Inder Kumar Gujral.
- **Activity 7 (model):** The first woman in space was Valentina Tereshkova; the first man to walk in space was Alexei Leonov; the first woman to walk in space was Svetlana Savitskaya; the first animal in space was Laika, the dog.
- **Activity 8 (present continuous):** (1) is studying; (2) are ... meeting; (3) is coming; (4) are playing; (5) is finishing; (6) is eating; (7) is not going; (8) am working; (9) are not cooking; (10) is walking.
- **Activity 9 (model):** see the model six-sentence night-sky paragraph on the StudyMatic chapter page.

Assessment & homework

- Oral: two fact-files compared; rapid-fire (dates, shuttles, the famous reply).
- Written: Activity 9 (six sentences on the night sky) + five Word Trove sentences.
- Digital: 10-question StudyMatic quiz on the chapter page (instant feedback).
- Project: make a poster — sketch and colour a space shuttle (Let's work together).

Free resource from studymatic.in · Built by Aftabuddin · This plan is an original teaching aid; it summarises but does not reproduce the textbook.